Advanced Data Science (ADS) Assignment

Total Marks: 20

- **Submission:** 10 marks
- **Viva Voce:** 10 marks
- **Last Date of submission**: (29th October, 2026)

Data-Driven Decision Making: Predictive Analytics and Causal Inference

This assignment explores two real-world-style decision problems using data science methods.

Part 1 focuses on predicting digital advertising clicks and making data-driven campaign recommendations.

Part 2 evaluates a job training programme using causal inference methods.

Follow the BADIR framework throughout: Business Understanding, Analysis, Data Insights, Inference, and Recommendations.

Part 1: Digital Advertising Click Prediction

Dataset: Dataset1.zip

Business Scenario

Digital advertising platforms display advertisements to users across different devices, applications, and websites. Understanding which impressions are more likely to generate clicks can help improve campaign targeting and advertising decisions.

1. Business Understanding

Define the business problem, identify the target variable, and formulate key analytical questions about advertisement click behaviour.

2. Dataset Summary and EDA

Use a summary tool to generate an overview of the dataset. Explain the important variables, their data types, meanings, and relevance to click prediction. Examine dataset dimensions,

missing values, class distribution, overall click-through rate (CTR), and CTR across selected device, app, and website features. Present visualisations and summarise key findings.

3. Machine Learning

Perform feature engineering and preprocessing. Build a baseline classification model, train the model, and compare their performance using appropriate evaluation metrics.

4. Inference and Interpretation

Interpret model performance and identify important predictive features. Explain the business implications and limitations of the findings. Distinguish predictive associations from causal relationships.

5. Recommendations

Propose evidence-based strategies for advertisement targeting and resource allocation. Support recommendations using findings from EDA and model evaluation.

Part 2: Evaluating a Job Training Programme Using Causal Inference

Dataset: jtpa_india_synthetic_10000.csv

Policy Scenario

A state skills mission randomly offered subsidised training to 10,000 applicants across 16 districts. Not everyone offered training enrolled, and a small proportion of applicants not offered training obtained it anyway. Employment and earnings outcomes were recorded at 18 and 30 months.

The government wants to understand the effect of offering training, the effect of training itself, and whether the programme benefits disadvantaged communities, including OBC, SC, and ST applicants.

Important: This is a synthetic dataset for a simulated policy evaluation exercise. Its results must not be presented as evidence about an actual Indian government programme.

Key Variables

Role	Variables
Assignment / instrument	`assigned_to_training`
Treatment received	`enrolled_in_training`, `training_type`, `months_in_training`
Outcomes	`employed_18mo`, `earnings_18mo_inr`, `employed_30mo`, `earnings_30mo_inr`
Baseline characteristics	`prior_12mo_earnings_inr`, `employed_prior_12mo`, `years_of_schooling`, `age_years`, `married`, `num_dependents`
Social groups	`social_category`, `gender`, `area_type`, `district`, `target_group`

1. Data Understanding and Randomisation

Examine the dataset structure, missing values, and descriptive statistics. Compare baseline characteristics between applicants assigned and not assigned to training. Assess whether randomisation appears successful and explain any observed imbalance.

Construct a 2 × 2 table of assignment and enrolment. Calculate enrolment rates and the first-stage effect. Explain noncompliance, monotonicity, and the exclusion restriction.

2. Intent-to-Treat (ITT) Analysis

Estimate the effect of assignment to training on earnings and employment at 18 and 30 months. Report point estimates, heteroskedasticity-robust standard errors, and confidence intervals. Repeat the analysis with baseline covariates and interpret the results for policymakers.

3. Instrumental Variables and LATE

Calculate the Wald estimator and estimate the effect of training on earnings using two-stage least squares (2SLS). Compare the IV and ITT estimates. Explain the roles of compliers, always-takers, and never-takers, and discuss the assumptions required to interpret IV as the Local Average Treatment Effect (LATE).

4. Earnings, Employment, and Heterogeneous Effects

Examine zero earnings and analyse the programme's effects on employment and total earnings. Estimate ITT effects separately for General, OBC, SC, and ST applicants. Present confidence intervals, discuss uncertainty, and explain the risks of multiple subgroup comparisons.

5. Inference and Recommendations

Based on the analysis, recommend whether the programme should be continued, expanded, or redesigned. Support your recommendations with numerical findings, discuss whether targeting particular groups is justified, and identify limitations and improvements for future evaluations.

Deliverables and Submission Guidelines

1. Submit a reproducible Python notebook or script for each part, or one notebook clearly divided into both parts.
2. Submit a concise report presenting the analysis, visualisations, model or causal estimates, interpretations, and recommendations.
3. Ensure all reported results are reproducible and supported by code.
4. Document preprocessing, sampling, modelling choices, evaluation methods, and relevant assumptions.
5. Avoid data leakage in predictive modelling and distinguish predictive association from causal effects.
6. For Part 2, explain the assumptions underlying causal estimates and distinguish statistical significance from practical and policy significance.